

Technical University of Munich | Practical course, extended to a paper | Supervisor: M. Sabanayagam

When Does Shift-Invariance Permit Adversarial Robustness?

A margin-to-Lipschitz account, from training-free geometry to AutoAttack

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Current status: technical-report paper at 11 pages with one central concept figure and four empirical regimes; a standalone proofs report at 23 pages; code and results pushed to branch `cifar-dissection`. Both documents compile cleanly and are kept in sync. This is a two-page summary for feedback, not a submission plan.

Summary. Whether shift-invariance helps or hurts adversarial robustness is contested: Ge et al. (2021) prove a fully shift-invariant classifier can be far less robust than a fully connected one, while anti-aliasing work reports the opposite correlation. We organise both regimes around one quantity. A shift-invariant linear classifier separates classes only through their projection onto the invariant subspace, with an exact margin identity (the distance between the projected convex hulls); a convolution-plus-pooling layer with a nonlinearity instead computes power-spectrum features, and a degree-three bispectrum invariant separates even the localized, phase-coded case that the power spectrum cannot. The surviving robustness is summarised by a margin-to-Lipschitz ratio η/L , a Lipschitz-margin certificate (no general converse) that also predicts trained-model robustness. A capacity-matched dissection on MNIST, Fashion-MNIST and CIFAR-10 confirms that η/L , not shift-consistency, predicts the adversarial robust radius (Pearson 0.998 on CIFAR), and decomposes each invariance operator’s effect into a margin term and a Lipschitz term: anti-aliasing acts on the Lipschitz term, exact invariance costs margin. Under PGD adversarial training a threat-matched η/L predicts AutoAttack robust accuracy (Pearson 0.90) while shift-consistency anti-predicts it (-0.88). On the optimization side we prove shift-invariance is margin-free on orbit-closed data, so gradient descent self-symmetrizes; it does help on near-orthogonal cluster-geometry data, where the trajectory-level selection theorem remains open.

1 Problem and motivation

Image classifiers are often called shift-invariant if circular translations of the input do not change the prediction. The published evidence on what this does to adversarial robustness is mixed. Ge et al. (2021) show a fully shift-invariant network needs an ℓ_2 perturbation of order $1/\sqrt{d}$ to be fooled, while a fully connected one needs order 1. Anti-aliased pooling work (Zhang 2019; Saha and Gokhale 2024) reports that operators which increase shift-invariance also increase robustness. A separate line proves a trade-off between geometric and adversarial robustness (Kamath et al. 2021; Tramèr et al. 2020). A practitioner is left without a rule for when invariance is safe.

What is already known. Invariant and equivariant architectures and their margins (Cohen and Welling 2016; Lawrence et al. 2021; Chen and Zhu 2023), invariant scattering and bispectrum features (Bruna and Mallat 2012; Kakarala 2012), Lipschitz-margin certification (Hein and Andriushchenko 2017; Tsuzuku et al. 2018), and implicit-bias non-robustness (Frei et al. 2023; Li et al. 2024) are all relevant pieces, but none ties shift-consistency to the adversarial radius through a single measurable quantity.

The gap. There was no account that (i) says exactly what discriminative signal survives a given invariance, (ii) turns that into a robustness certificate, and (iii) is validated against a standardized strong attack at matched capacity. This project supplies all three.

2 Method

Structural core. For a finite symmetry group acting orthogonally, an invariant linear classifier sees the data only through the group-average projector P_G ; its maximum ℓ_2 margin is $\frac{1}{2} \text{dist}(\text{conv } P_G X_+, \text{conv } P_G X_-)$ (the

cyclic-shift case is the DC projection). A single convolution, square, and global-average-pooling layer computes power-spectrum coordinates; a degree-three bispectrum invariant additionally separates phase-coded classes such as the Ge black/white dot $\pm e_j$ that the power spectrum cannot.

The criterion. The surviving robustness is summarised by η/L : the margin η of the data in the invariant feature space divided by the Lipschitz constant L of that feature. The bound $r_2 \geq \eta/L$ is a one-sided Lipschitz-margin certificate; we show there is no general converse, so η/L is a certificate and a predictor, not a characterization. A companion orbit-flip radius ρ_G bounds the excessive-invariance (semantic) side, recovering Tramèr et al.

Experiments. A training-free test computes the achievable separation by classifier family. A capacity-matched dissection trains four architectures (standard, anti-aliased, exact cyclic, shift-augmented) with identical parameter counts, measuring the per-sample ℓ_2 robust radius with a decoupled direction-and-norm minimum-norm attack (validated against the closed-form linear radius to under one percent) and the margin/Lipschitz decomposition. A PGD-adversarial-training arm makes AutoAttack informative. A synthetic optimization study compares unconstrained and weight-tied ReLU networks on orbit data.

3 Main empirical evidence

The training-free test confirms the structural predictions exactly: the single dot is separable by an unconstrained classifier, weakly by linear invariance, not at all by the power spectrum; orthogonal-frequency data is separable by neither linear classifier yet separable by the power-

spectrum invariant.

η/L predicts robustness; shift-consistency does not. Across capacity-matched cells, η/L tracks the robust radius (Spearman 0.96 on MNIST, 0.71 on Fashion-MNIST; Pearson 0.998 on CIFAR-10), while shift-consistency is uncorrelated or mildly anti-correlated. The exactly invariant model has shift-consistency 1.0 yet the smallest robust radius.

CIFAR-10 arm ($w=64$)	consist.	r_2	η/L
standard	0.89	0.119	0.117
anti-aliased	0.91	0.141	0.144
exact cyclic	1.00	0.090	0.088
shift-aug	0.94	0.117	0.117

Identical parameter counts per width. The most shift-consistent arm (exact cyclic) is the least robust; anti-aliasing is the most robust.

Under adversarial training. PGD-AT at $\ell_\infty=8/255$ makes AutoAttack informative (27–41% robust accuracy, no gradient masking). Shift-consistency still anti-predicts robustness (Pearson -0.88), and the threat-matched ratio $\eta/\|\nabla M\|_1$ (the ℓ_∞ dual) predicts AutoAttack robust accuracy (Pearson 0.90), while the ℓ_2 ratio does not, because adversarial training robustifies in ℓ_∞ .

4 Mechanism diagnostics

The margin/Lipschitz decomposition separates two mechanisms. Relative to the standard baseline, anti-aliasing leaves the margin essentially unchanged and lowers the Lipschitz constant ($\Delta \log L \approx -0.2$), buying robustness through reduced input sensitivity. Exact cyclic invariance instead shrinks the margin ($\Delta \log \eta \approx -0.3$ to -0.4) and loses robustness, the real-data form of the Ge collapse. This is the structural criterion made operational: invariance helps robustness only when it does not cost margin.

Optimization. On orbit-closed data we prove shift-invariance is *margin-free*: the invariant and unconstrained two-layer-ReLU max-margin values coincide (orbit-averaging an unconstrained solution gives an invariant one of equal minimum norm). A synthetic test confirms unconstrained and tied networks reach the same robustness on generic orbit data, while on near-orthogonal cluster-geometry data the tied network reaches a 20–60% larger η/L , most in the lazy regime. So whether invariance re-points the implicit bias is data-geometry-dependent.

5 What was already done

- **Structural theory**, corrected after an external audit and re-verified: invariant margin as projected-hull distance (general finite group), the η/L certificate with an explicit no-converse counterexample, the power-spectrum and bispectrum span lemmas, the exact orbit-rank lemma, and the orbit-flip bound.
- **Capacity-matched dissection** on MNIST, Fashion-MNIST and CIFAR-10; the robust radius validated against AutoAttack to a mean gap of 0.014, ruling out

gradient masking.

- **PGD adversarial-training arm** with AutoAttack and the threat-matched η/L law.
- **Optimization sprint**: a margin-free theorem on orbit-closed data, a synthetic unconstrained-versus-tied study showing the data-geometry dichotomy, and explicit ReLU KKT systems for the open selection problem.
- **One central concept figure**; two documents kept in sync; code and results on branch `cifar-dissection`.

6 Future work

- **Cluster-geometry selection theorem.** Prove that on near-orthogonal cluster-geometry orbit data gradient flow selects a higher- η/L invariant solution, using the explicit KKT systems and the frequency-domain reduction; the lazy regime shows the largest empirical gap.
- **Bispectrum certified robustness.** Turn the bispectrum separation of phase-coded classes into a certified-radius construction, trading the larger Lipschitz constant against the recovered margin.
- **Scale.** Larger backbones and stronger adversarial-training recipes, to check whether the threat-matched η/L law and the consistency anti-prediction persist at RobustBench scale.

7 Project context

This grew out of a TU Munich practical course that reproduced and extended Ge et al. (2021). The supervisor’s framing asked for a comparison against four papers: Ge et al. (2021), the two Kamath et al. trade-off papers, and Galloway et al. on batch normalization; the dissection reproduces the Galloway effect under fixed- ϵ accuracy and shows it is threshold-dependent under the radius metric. Repository: `adversarial-robustness-shift-invariance/`; paper and proofs report under `paper/report/`. Compute to date is a few GPU-hours on two RTX A6000.

Selected references. Ge, Singla, Basri, Jacobs, Shift Invariance Can Reduce Adversarial Robustness, NeurIPS 2021. Kamath, Deshpande, Subrahmanyam, Balasubramanian, Can We Have It All?, NeurIPS 2021. Tramèr, Behrmann, Carlini, Papernot, Jacobsen, Fundamental Tradeoffs Between Invariance and Sensitivity, ICML 2020. Galloway et al., Batch Normalization Is a Cause of Adversarial Vulnerability, 2019. Zhang, Making Convolutional Networks Shift-Invariant Again, ICML 2019. Croce, Hein, Reliable Evaluation with AutoAttack, ICML 2020. Rony et al., Decoupling Direction and Norm for Min-Norm Attacks, CVPR 2019. Frei, Vardi, Bartlett, Srebro, The Double-Edged Sword of Implicit Bias, COLT 2023. Lyu, Li, Gradient Descent Maximizes the Margin of Homogeneous Networks, ICLR 2020. Lawrence, Georgiev, Dienes, Kiani, Implicit Bias of Linear Equivariant Networks, ICML 2022. Chen, Zhu, On the Implicit Bias of Linear Equivariant Steerable Networks, NeurIPS 2023. Kakarala, The Bispectrum as a Source of Phase-Sensitive Invariants, 2012.